



Towards a Framework of Process Owner Competencies and Tasks

Alfred Rudman

University of Cape Town
Department of Information Systems
+27 (0)21 650 4259

alfredrudman@gmail.com

Malcolm Garbutt

University of Cape Town
Department of Information Systems
+27 (0)82 445 6405

Malcolm.Garbutt@alumni.uct.ac.za

Lisa F Seymour

University of Cape Town
Department of Information Systems
+27 (0)21 650 4259

Lisa.Seymour@uct.ac.za

ABSTRACT

Process owners are vital to the establishment and functioning of process oriented organisations. However there is a paucity of understanding regarding the tasks process owners should undertake and what competencies they require. Sets of process owner competencies and process owner tasks emerged from interviews with executives from three financial services organisations in South Africa. The first set highlighted business process management, interpersonal, and holistic thinking competencies while from a tasks perspective process measuring, customer experience, staffing, and process improvement were highly ranked. Nonetheless process orientation was found to be more aspirational than applied which was ascribed in part to a lack of business process education.

CCS Concepts

• Applied computing → Business process

Keywords

Business process monitoring; process owner; process orientation; process owner competencies; process owner tasks.

1. INTRODUCTION

Business process management (BPM) can assist organisations in sustaining competitive advantage in the marketplace through embracing and implementing best practice management principles, strategies and technologies on a continuous basis [1]. BPM is defined as a management practice that integrates business process knowledge and information technology with the aim of transforming business efforts into integrated and measurable cross-functional activities that deliver strategic and operational competitive advantage [2]. Moreover it holds promise for gaining competitive advantage through continuous process improvement when organisations adopt a process orientation approach [1]. The importance of managing and improving corporate components including process strategy, process architecture, process ownership, process measurement and process improvement [3] are recognized by many organisations. However managers of adopting organisations need to understand the implications and

impact of process adoption [1]. Efforts to improve business processes have historically been incomplete focusing on functional automation of processes and utilising information technology. Exclusively automating processes has limited impact on competitive position due to ignoring end-to-end cross-functional linkages which are important from a customer retention and satisfaction perspective [4]. This leads to reduction in end-to-end process control and limited process ownership [5] with non-evolving processes and lack of improvement opportunities. Consequently the appointment of process owners and the implementation of process performance monitoring for process improvement resulting from process initiatives is recommended [1].

While the appointment of process owners is an important aspect in BPM [6] there is a paucity of empirical studies investigating exactly what the process owner role and tasks entail in practice [7]. According to Nesheim [7] tasks are based on primary competencies that should be augmented through learning from specific situations. The objective of this research paper is to identify process owner tasks and the primary competencies necessary to facilitate effective BPM.

This paper is organised as follows. In section 2 a review of the literature is presented prior to an overview of the research methodology in section 3 and a summary of the findings in section 4. Section 5 presents a discussion of the findings precedes the conclusion in section 6.

2. LITERATURE REVIEW

Successful BPM implementations necessitate adoption of a process oriented view of the organisation. Successful transition to a process oriented organisation entails appointing process owners, establishing BPM and creating governance boards [4]. BPM governance is required to align systems and practice [8] for which organisational design changes are necessary [6] and consequently management. Continuous process improvement through (re-)alignment with the organisational environment is vital for benefits from BPM [9]. Such process improvement must be performed in an active and systematic manner [1]. While process improvements have the ability to increase staff satisfaction they have been observed to be critical in driving customer satisfaction especially in the financial services sector [10] leading the researchers to conclude that *“practitioners should focus on process management ... rather than simply addressing service quality from a functional perspective”*.

Process owners are the most visible difference between process oriented and traditional organisations [11] and are crucial to the effectiveness of process orientation [3], [12]. They are viewed as the champions of process and take overall responsibility for process performance being central in interfacing cross-functional processes [13]. Process owners should be appointed on a full time

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from Permissions@acm.org.

SAICSIT '16, September 26-28, 2016, Johannesburg, South Africa

© 2016 ACM. ISBN 978-1-4503-4805-8/16/09...\$15.00.

DOI: <http://dx.doi.org/10.1145/2987491.2987522>

basis to assist in evolution of processes and to curb the organisation reverting to less effective processes [11]. Hence identification and allocation of process owners to the core business processes is a key element for effective BPM [10]. Nevertheless there is a paucity of empirical research on what process owner should be doing on a day-to-day basis [1], [7]. The fundamental role that a process owner plays entails the management of interfaces between key processes with the objective of preventing horizontal silos. The process owner has overall responsibility of the end-to-end process performance utilising metrics to track, measure and monitor the status of the business process with emphasis on continuous improvement initiatives [1]. Associated with process performance monitoring is capacity planning which requires process owners to ensure that staff are adequately trained and allocated to processes. In practice this may result in conflicts arising between process owners, teams, and functional management [3].

Whereas the majority of organisations with BPM programs assign process owners BPM programs are found frequently to be either ad-hoc or project-based [12] and consequently without dedicated process owners. While process owners are considered responsible for managing processes little is known about what they are supposed to do beyond responsibility (a) for process performance and (b) for process improvement [12]. In the absence of a specific process owner task framework general BPM tasks may be applicable to the process owner role. A literature review, for instance, found three groups of BPM tasks which bear relationship to process owners: designing processes; day-to-day management of processes; and fostering process related learning [14]. However this postulation is not supported empirically. Furthermore limited knowledge of the tasks that process owners should be performing on a daily basis indicates that the competencies required by process owners in performing their tasks are also unidentified. Process owner competencies are understood as the knowledge, skills and attitudes required by process owners to perform business process tasks effectively [15].

Consequently the aim of this research is to explore what tasks process owners should be undertaking and what supporting competencies they require to enact such tasks.

Identifying the competencies and tasks of process owners has implications for practitioners and academia. It can provide practitioners with a checklist when appointing and monitoring process owners while affording understanding of the significance of the process owner role. Academically this research supplies empirical evidence of the competencies required by process owner and the tasks they should be performing.

Two research questions guided the research:

1. What business process competencies are required by business process owners?
2. How are these competencies enacted in practice?

3. RESEARCH METHODOLOGY

Research was undertaken at three financial services organisations in Gauteng, South Africa with whom the first author has a business relationship. A qualitative, exploratory, and interpretive study in a cross-sectional timeframe using semi-structured interviews was followed. The exploratory, interpretive research philosophy was adopted for this study to build theory through subjective interpretation of participants' perspectives [16]. The use of exploratory research is especially useful for investigating a new area of inquiry in order to determine the feasibility of further research with regards to that phenomenon [16].

Primary data took the form of semi-structured interviews which were analysed in accordance with the general inductive theory proposed by Thomas [17]. The interviews were recorded and transcribed prior to reading and re-reading. Codes were then assigned to relevant portions of the transcriptions and aggregated into thematic categories. The categories and revised codes are presented in the next section.

4. FINDINGS

Eleven respondents from three South African financial services companies were interviewed over a period of two months in 2014. Four executives, six senior managers and one consultant participated in the research.

From the interviews three competency categories and three tasks categories emerged as shown in Table 1. The three Competency categories comprise task related competencies; people related competencies; and thinking competencies. The three Tasks categories comprise Planning tasks; Controlling tasks; and Governance tasks.

Table 1: Competency and Task Categories Categorised into Concepts

Categories	Concepts
Competencies	Task related competencies
	People competencies
	Thinking competencies
Tasks	Planning tasks
	Controlling tasks
	Governance tasks

4.1 Process Owner Competencies

Competencies in the form of knowledge, skills and, to a lesser extent, attitudes emerged from the data within the three categories as shown in Table 2. The most significant competency in the study was the task of business process management (count = 79). This was followed by competencies for customer focus (count = 10), change management (count = 9) and governance (count = 5). The second most important set of competencies relates to people competencies (count = 34) comprising interpersonal competencies (count = 26) and organisation understanding (count = 8). The third competence identified was systemic thinking (count = 19).

Overall the highest ranked individual competencies required by business process owners were found to be business process management (count = 79), interpersonal competencies (count = 26), systemic thinking (count = 19) and customer focus (count = 10).

Table 2: Process Owner Competencies

Category	Competency	Count
Task	Business Process Management	79
	Customer Focus	10
	Process Change Facilitation	9
	Process Governance	5
People	Interpersonal	26
	Organisation Understanding	8
Thinking	Systemic Thinking	19

The competencies are reviewed and supported by extracts from the interviews in the rest of this section.

4.1.1 Process Owner competencies: Task Related

Task related competencies emerged from four task related categories: (a) business process management (count = 79); (b) customer focus (count = 10); (c) process change facilitation (count = 9); and (d) process governance (count = 5).

Business process management competency requirements were observed to be a balance between understanding and practical application of business process management activities. Specific understandings identified were (i) process design and improvement methodologies, tools and techniques (count = 45), (ii) process management principles (count = 8), (iii) business process management frameworks, standards, and implementation (count = 8), (iv) numerical and data analytical skills (count = 6), (v) technology architecture supporting business processes (count = 4) (vi) process mapping standards (count = 3), (vii) implementation and monitoring of process metrics and measurements (count = 3), and (viii) work demand and staff capacity planning requirements (count = 2).

Respondent IP4 encapsulated task related competencies as: “... identify all the relevant processes ... to monitor the process ... too much capacity, do you have too little capacity ... how does technology enable ... based on people, process and technology. People and technology will support the process ... legislation and compliance ... need to understand it, they need training on it ... to understand the importance of following the process, in alignment with our customer value proposition ... to think out of the box ... able to motivate ... to persuade people down the right path ...”

4.1.2 Process Owner competencies: People

People competencies are categorised as (a) interpersonal (count = 26) and (b) organisational understanding (count = 8).

Interpersonal competencies emerging from the interviews included: (i) influence and negotiation with personal mastery (count = 10); (ii) people management and motivation (count = 7); (iii) communication (count = 4); (iv) leadership and management (count = 3); and (v) creative and innovative thinking (count = 2).

According to respondent IP3: “... problem solving skills, and interpersonal skills ... to be able to communicate effectively, because there is a reliance on other people to effect the process as intended. Innovation skills. Effective planning skills ...”

At the same time process owners must understand organisational context, culture and principles of organisational design.

IP11: “... you need a good understanding of the business, business model, the whole value chain”

4.1.3 Process Owner competencies: Thinking

Systemic thinking (count = 19) emerged as the third set of task related competencies. It was ranked second of all the individual competencies below BPM methodologies and tools.

IP2: “... able to apply systemic thinking... the ability to not just look at things in isolation ...”

4.2 Process Owner Tasks

Three clusters of process owner tasks emerged from the research: planning (count = 196); control (count = 68); and governance (count = 15). Similarities were observed between competency findings and task findings as shown in Table 3.

Table 3: Process Owner Tasks

Tasks	Planning	Control	Governance
Process Measurement	45	48	
Customer Experience	43		
Process Change	34	16	10
Governance	24		
Capacity	19	4	
Staff Training	15		2
Technology	7		
Alignment	6		
Change Management	3		3

4.2.1 Process Owner tasks: Planning

Nine categories of planning tasks were observed: (a) process measurement (count = 45); (b) customer focus (count = 43); (c) business process change (count = 34); (d) governance (count = 24); (e) capacity planning (count = 19); (f) staff training (count = 15); (g) technology (count = 7); (h) strategic alignment (count = 6); and (i) change management (count = 3).

Process measurement planning activities included: (i) identifying, designing and implementing process metrics and measurements (count = 26); (ii) implementing process metrics reporting mechanisms (count = 8); (iii) identifying and implementing staff process key performance indicators (count = 7); (iv) conducting time and motion studies (count = 3); and (v) aligning staff remuneration with improvement agenda (count = 1).

IP2: “... ensuring that the line functions have appropriate measurements ... and reinforcing that by metrics and incentives ... Constantly testing what you have done with the market ...”

Four customer focused planning tasks were identified to: (i) design customer-focused interaction mechanisms (count = 14); (ii) design processes in line with customer expectations (count = 13); (iii) identify customer process performance expectations (count = 11); and design customer entry-points and transactions (count = 5).

IP3: “... we want to improve customer experience and ... design now will identify at what points we push and pull feedback ... how that alignment has come through ...”

Five specific business process change tasks were identified: (i) identify incremental and re-engineering process enhancement opportunities (count = 18); (ii) design processes with minimal process hand-offs (count = 10); (iii) create and execute process improvement plans (count = 2); (iv) identify process automation opportunities (count = 2); and (v) design pilot process to prove performance meeting abilities (count = 2).

IP2: “... where we going to have incremental effort, where we going to have major re-engineering effort ... Almost like a journey plan ... Test what the customers are experiencing in the new process ...”

Ranked fourth of the planning categories but third as an individual task was governance in line with corporate policies and standards (count = 24). Controlling and planning of metrics was ranked slightly higher than governance followed by capacity planning.

IP3: "... standards and procedures ... must be maintained and reviewed periodically to ensure they are aligned ..."

Capacity planning (count = 19) was ranked fifth for planning tasks and fourth as an individual task.

IP6: "... to measure the data on a daily basis, what's there, what the demand, what work was left over, how many people they've got ..."

Two aspects of staff training were identified: (i) development of organisational knowledge (count = 9); and development of business process skills (count = 6) which speaks more to education from an academic perspective.

IP4: "... they need to understand it, they need training on it, they need to understand what the repercussions or impacts are on the process, they need to understand the importance of following the process ..."

The seventh task was identified as the determination of technology requirements in line with process objectives (count = 7).

IP4: "... how does technology enable ... technology will support the process ..."

Strategic alignment of business processes with business objectives (count = 6) emerged as the eighth planning task of business process owners.

IP4: "... a systematic approach to align an organisations process ..."

The ninth and last planning task of process owners was identified as change management (count = 3).

IP5: "... using change management process ... with diplomacy rather than forced fit ..."

4.2.2 Process Owner tasks: Controlling

The second process owner task category emerged as (a) controlling process metrics (count = 48), (b) business process change (count = 16) and (c) capacity planning (count = 4).

Process metrics monitoring emerged as the primary controlling task. Five activities were identified: (i) monitor process performance (count = 31); (ii) provide process performance reports (count = 7); (iii) monitor customer satisfaction and process expectations (count = 4); (iv) monitor process performance metrics (count = 3); (v) manage process performance exceptions (count = 3).

IP6: "... monitor them, understand if there are exceptions, what the exceptions are ... and take necessary action. Continuous monitoring, understanding and questioning ..."

Business process change controlling tasks were found to comprise three activities: (i) identify process improvement opportunities (count = 12); (ii) manage process hand-offs (count = 3); (iii) monitor customer retention (count = 1).

IP7: "... the first objective would be to just reduce hand-offs dramatically ... then do hand-offs ... in a controlled and result based mind-set ..."

Monitor capacity planning (count = 4) was ranked third of the controlling tasks for process owners.

IP4: "... also look at capacity ... because that impacts our processing time ..."

4.2.3 Process Owner tasks: Governance

The final task category was governance (count = 15) comprising (a) governing process change (count = 10), (b) process exception management (count = 3) and (b) documentation for staff process training (count = 2).

Two tasks emerged for governing process change: (i) review and maintain processes (count = 9); and (ii) conform to process documentation standards (count = 1).

IP3: "... the process needs to be documented ... standards and procedures and those must be maintained and reviewed periodically ..."

The findings observed in the study and summarised above are discussed in the next section.

5. DISCUSSION

Three competency categories and three tasks categories emerged from the findings. Competencies and tasks were observed to be entwined with overlapping concepts inter-category and intra-category. This is in accordance with the view of competencies as knowledge, skills and attitudes required for performing tasks [15].

5.1 Process Owner Competencies

The highest ranked competencies for a business process owner were observed to be knowledge and skills in business process management followed by interpersonal skills and the ability to think systemically. While the competencies were logically grouped into task-related, people-related, and systemic thinking categories they exhibited considerable interaction between them.

Although empirical research into process owner competencies are limited and little is known about what process owners should be doing [12] the findings are in line with other research into competencies of business process practitioners. In a South African study into competencies of enterprise resource planning consultants qua process oriented practitioners, BPM competencies were observed to be the highest ranked technical competencies. However overall interpersonal skills were ranked higher followed by business knowledge [18]. In the current study business process management was ranked higher than interpersonal skills. Further differences between process owners and enterprise resource planning consultants included systemic thinking and the lower ranked competencies of focus on the customer, facilitating business process change, and process governance.

In this study, some respondents showed concern that process owners wield too much power in the organisation which could lead to interpersonal conflict. In prior studies, interpersonal skills have been found to be significant and although competence and trust in process owners were deemed to be high, communication skills were ranked moderately lower [7]. Consequently it was recommended that process owners communicate and engage in real dialogues that builds relations [7].

Systemic thinking is a significant competency of process owners who need to be creative in process design [7]. They must understand the process in totality including boundaries and inter-relationships [3]. A holistic view combining strategy, operations, people, techniques and people with alignment to corporate objectives, governance and customer focus are important for adequate business process management [19]. Likewise meeting the needs of customers is at the core of business processes [3]. In the conclusion of a quantitative study in the financial services industry the importance of BPM in "driving customer satisfaction" is emphasised [10].

Process change necessarily impacts on the design of organisations [7] which must be prepared for cultural changes [3]. Such change must be addressed from the process perspective [3] and properly governed. Ongoing governance must also support a change management system in affecting process changes [3]. A definitive link exists between customer focus, alignment and governance [19].

5.2 Process Owner Tasks

Three clusters of process owner tasks (planning, control, and governance) emerged with significant overlap between individual concepts both inter-task and between tasks and competencies. Highest ranked process owner tasks were observed to be metrics control and planning, customer experience planning and process change planning. Process change was highly ranked for control and governance while capacity planning was also ranked highly.

Process owners track business process performance through metrics which are used as guides for improvement [11]. Consequently process measurement is extremely important for performance management and integral to BPM [6]. The aim of measurement is optimisation of process performance from both economic and customer requirements perspectives [3]. However these measurements must be balanced and aligned to organisational strategy with the aim of process improvement [6]. This finding empirically confirms the close cooperation between appointing process owners and process performance measurement which should be implemented in tandem [1]. However poor process design may cause a performance ceiling whereby the effectiveness of processes is limited. Highlighted in this finding is the importance of appointing process owners [1] and the need for process improvement [3], [20]. This is supported by the findings in a research setting similar to the current study where the most important tasks of process owners were observed to be 'monitoring process improvement', 'problem solving' and 'initiating process improvement' [12].

While customer satisfaction is important [19] it has been shown to be a low success factor in business process management [21]. Incorrect focus on measurement of customer requirements [3] rather than customer satisfaction or experience may be the reason for this finding. This is exemplified in the view that organisational design and internal issues are key to BPM success [7]. Consequently customer satisfaction should be part of the process measurement mix which drives process change and improvement. Process improvement meanwhile is observed in forms, (a) as improvement of business processes [3], [20], and (b) improvement in the process approach evolving from modelling to process improvement [21]. A matter brought to light in this study and recognized as being of particular importance is the management of hand-off boundaries between processes to ensure sustainable organisational performance and meeting customer and stakeholder expectations [6].

Process change may be accomplished either incrementally or through radical changes to business processes [6]. Consequently process owners must constantly be aware of process improvement opportunities in addition to their process management obligations [12].

A further low success factor of BPM according is governance according to [21] who in a later paper quoted in [8] defines BPM governance as: process management decision making; process roles and responsibilities; process metrics and performance linkages; process-related standards; and process compliance. Governance is crucial in business process management which aims at standardisation processes from the ideology of 'one-best-

way' [7]. However process owners have been observed to become obsessed with compliance which may lead to interpersonal conflict. A further concern is the practice of governance being limited to "keeping the process documentation up-to-date" [12].

A source of dispute may also arise from the vital task of staff capacity planning in trying to provide a balance between optimal process performance and optimal organisational performance [7]. Process owners need to instill trust and prove efficiency in distributing resources. Although some authors focus on incremental change [7] others focus on resource allocation during process improvement projects to ensure process compliance [5], [12]. Linked to staffing is staff training. Process owners must ensure that staff allocated to the processes are suitably trained and understand the process they perform [20]. Such training must address staff competence at both the individual and collective levels from a strategic perspective [7].

Technology is vital to BPM [2] and together with strategic alignment, governance, methods, people, and culture is the core of BPM [22]. Analogous to the current research technology has been observed as a low ranked competence amongst process oriented practitioners [18]. A further low success factor of BPM is alignment [21] nonetheless it is considered vital to a comprehensive business process management approach [19] and one of the six core elements of BPM [22].

Likewise while change management was not ranked high in this study it is essential in supporting process governance and maintaining an ongoing representation of processes [3]. It may however be tempered by lack of readiness of the organisation to accept cultural change' [3] which will impede process improvement [9].

5.3 Summary

While process owner competencies and tasks are poorly defined in the literature [7] the findings from this study show that practitioners' understandings are in accordance with the business process management requirements from early days of the field. Armistead [13] provided ten principles of managing business processes: (i) designate a process champion; (ii) know the process (iii) understand the linkages; (iv) work on the trade-offs; (v) teach others about the process; (vi) train within the process; (vii) measure the process; (viii) manage careers; (ix) build specialist expertise; and (x) improve the process. Three groups of tasks have been suggested: designing processes, day-to-day process management; and nurturing process related learning [14]. Over time technology has become more relevant while interpersonal skills have become vital [18]. In addition a need for a stronger focus on a holistic view and alignment of strategy, operations, and people to corporate objectives, governance and customer focus has emerged [19]. All of these tasks are shown to be relevant to practitioners' belief of process owner competencies and tasks. However due to the paucity of dedicated process owners amongst the respondents evidenced by the aspirations of business for a process management solution this may be more theoretical than based on experience. By virtue of the current study involving only financial services organisations within a particular geographic region cognisance must be taken of differences between BPM implementations in the manufacturing sector and the services sector. Business process focus has been found to be significantly lower in service sector organisations than in manufacturing sector organisations [4]. This resonates with the low level of process focus observed in the study as shown in the following excerpts from the interviews. *IP6: "... but the processes ... are still working in silo's."* *P3: "... the processes enable us to manage*

work, as it suits us but they have not been design[ed] with the customer in mind ...”

Whereas the process owner should be a permanent position [11] in the case organisation process management was found to be performed intermittently: IP4: “...on an ad-hoc basis, as and when we need it.”

Likewise the appointment of process owners was observed to be limited: IP1: “*The one principle we have agreed on is that processes can’t be owned by a process custodian or process engineer, it must be owned by the people who do the work every day. The business.*”

Nonetheless most of the respondents were found to aspire to implementing business process initiatives. IP5: “... currently utilising, workflow discussions ... Our IT direction, is taking all those process discussions into consideration ...” IP7: “... We’ve selected the framework. We haven’t actually created the capability” IP4: “... we’re having workshops, we’re doing the processes, we’re going to test the processes”

5.4 Implications for Practice

The study provides advice and warnings for organisations who aim to become process oriented. First dedicated cross-functional process owners who are responsible and held accountable for process measurement and improvement should be appointed. These process owners must align corporate strategy with customers focus and have sufficient authority to make governable changes across the organisation. They must be process champions able to equip the process with educated staff while holistically observing improvement opportunities beyond the restrictive scope of a formalised task framework.

6. CONCLUSION

This study set out to explore the competencies required by process owners and the application of these competencies in the form of tasks. Based on an understanding of the tasks and required competencies of process owners identified in business process literature a study was conducted at three financial services organisations in Gauteng, South Africa. Following a general inductive approach it was observed that process owner competencies and tasks are in accordance with universal BPM task requirements identified internationally over the past two decades. The study also found a low level of business process focus in the services industry in accordance with the literature. However aspirations to implement business process management systems was evident with projects already initiated.

The paper contributes to the business process management body of knowledge in highlighting the importance of systemic thinking and customer focus in addition to BPM knowledge and interpersonal competencies. From a tasks perspective the contribution to current knowledge is seen primarily in the acknowledgement of the importance of process performance measurement. Further vital tasks were found to be customer focus driven process improvement and governance.

However a limited use of business process management was observed which was partially ascribed to a lack of adequate process owner / process management education programs. IP1: “... benefit from such training programs? Without a doubt. One of the problems you have in most of the business units is that they don’t have process engineers ...”

Education is vital to process owners as research has observed them doing “*different things than what they are supposed to do*”

[12]. This means that managers must also be suitably educated in process management in order to ensure that the practices perform appropriately [1]. Future research may assist organisations in the development of education programs, which would improve maturity levels of process owners from a competency and task performance perspective.

Limitations of the study arise from having only 11 respondents from 3 financial services organisations in a single area. Furthermore the respondent companies all had limited BPM initiatives and no permanent process owner role. Future research using financial services with established business process management implementations may provide a richer picture of the process owner tasks and required competencies.

7. REFERENCES

- [1] Kohlbacher, M., and Gruenwald, S. 2011. Process ownership, process performance measurement and firm performance. *International Journal of Productivity and Performance Management*, 60, 7, 709–720.
- [2] Antonucci, Y. L., and Goeke, R. J. 2011. Identification of appropriate responsibilities and positions for business process management success: Seeking a valid and reliable framework. *Business Process Management Journal*, 17, 1, 127–146.
- [3] Smart, P. A., Maddern, H., and Maull, R. S. 2009. Understanding Business Process Management: Implications for Theory and Practice: Understanding Business Process Management. *British Journal of Management*, 20, 4, 491–507.
- [4] Goeke, R. J., and Antonucci, Y. L. 2013. Differences in Business Process Management Leadership and Deployment: Is There a Connection to Industry Affiliation? *Information Resources Management Journal*, 26, 2, 43–63.
- [5] Larsen, M. H., and Klischewski, R. 2004. Process ownership challenges in it-enabled transformation of interorganizational business processes. In *System Sciences, 2004. Proceedings of the 37th Annual Hawaii International Conference on*, 1–11. Hawaii: IEEE.
- [6] Armistead, C., Pritchard, J.-P., and Machin, S. 1999. Strategic business process management for organisational effectiveness. *Long Range Planning*, 32, 1, 96–106.
- [7] Nesheim, T. 2011. Balancing process ownership and line management in a matrix-like organization: Balancing process ownership and line management. *Knowledge and Process Management*, 18, 2, 109–119.
- [8] Doebeli, G., Fisher, R., Gapp, R., and Sanzogni, L. 2011. Using BPM governance to align systems and practice. *Business Process Management Journal*, 17, 2, 184–202.
- [9] Niehaves, B., Poeppelbuss, J., Plattfaut, R., and Becker, J. 2014. BPM capability development – a matter of contingencies. *Business Process Management Journal*, 20, 1, 90–106.
- [10] Kumar, V., Smart, P. A., Maddern, H., and Maull, R. S. 2008. Alternative perspectives on service quality and customer satisfaction: the role of BPM. *International Journal of Service Industry Management*, 19, 2, 176–187.
- [11] Hammer, M., and Stanton, S. 1999. How process enterprises really work. *Harvard Business Review*, 77, 1, 108–120.

- [12] Reijers, H. A., and Peeters, R. L. 2010. Process owners in the wild: Findings from a multi-method descriptive study , 1–12. Presented at the ER-POIS 2010, Hammamet, Tunisia.
- [13] Armistead, C. 1996. Principles of business process management. *Managing Service Quality: An International Journal*, 6, 6, 48–52.
- [14] Paim, R., Mansur Caulliraux, H., and Cardoso, R. 2008. Process management tasks: a conceptual and practical view. *Business Process Management Journal*, 14, 5, 694–723.
- [15] Ravesteyn, P., Batenburg, R., and de Waal, B. 2008. In search of competencies needed in BPM projects. *Communications of the IIMA*, 8, 2, 23–30.
- [16] Bhattacharjee, A. 2012. Social science research: principles, methods, and practices.
- [17] Thomas, D. R. 2006. A general inductive approach for analyzing qualitative evaluation data. *American Journal of Evaluation*, 27, 2, 237–246.
- [18] Scholtz, B., Cilliers, C., and Calitz, A. 2011. Critical competencies for South African ERP consultants, 1–25. Presented at the International Business Conference, Curepipe, Mauritius.
- [19] De Bruin, T., and Rosemann, M. 2005. Towards a business process management maturity model. In *ECIS 2005 Proceedings of the Thirteenth European Conference on Information Systems*, 1, 1–12. Regensburg, Germany.
- [20] Hammer, M. 2002. Process management and the future of six sigma. *Sloan Management Review*, 43, 2.
- [21] De Bruin, T. 2007. Insights into the evolution of BPM in organisations. In *ACIS 2007 Proceedings*, 43, 632–642. Toowoomba, Australia: Australasian (ACIS).
- [22] Rosemann, M., and vom Brocke, J. 2015. The six core elements of business process management. In *Vom Brocke, J., Roseman, M. Handbook on business process management 1: Introduction, methods, and information systems. (2nd Edition)*. New York: Springer.